

Exoskeletons

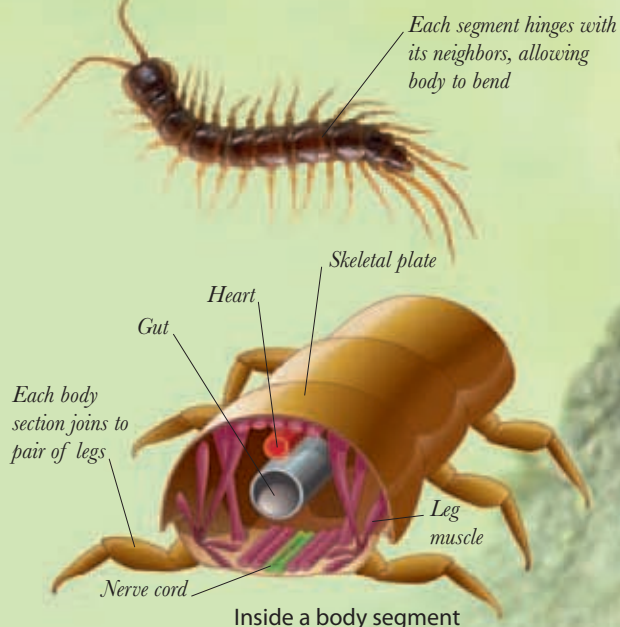
Arthropods, including insects, spiders, crustaceans, and centipedes, make up more than four-fifths of the world's animals. All of these creatures have an exoskeleton—an outer body case with flexible joints. Exoskeletons are built from chitin, a substance that resembles a natural plastic, together with minerals that provide extra strength. Exoskeletons are extremely good at withstanding wear and tear. Their main disadvantage is that they cannot enlarge once they have formed. To get around this problem, arthropods shed, or molt, the outer layer of their exoskeletons as they grow.

Armored protection

Brandishing their huge, antlerlike jaws, or mandibles, two male stag beetles prepare to do battle. As with all insects, these rival beetles have an exoskeleton made of tough chitin plates with joints that allow their bodies to move. The hardened forewings, called elytra, cover and protect their transparent hind wings and their large mandibles are hollow, which saves on weight. In spite of all this body armor, stag beetles fly well, although not very fast.

Body segments

An arthropod's body is divided into jointed segments. In centipedes, these segments are repeated along the length of the body. Each one is surrounded by hard body plates. These provide anchorage for the centipede's internal muscles, including those that reach into the legs.



Chest, or thorax, casing protects wing and leg muscles

Hardened forewings, or elytra, protect rear body, or abdomen

Tough plates fused together form boxlike head

Powerful mandibles used to fight rival males

Tubular leg with five joints